

Disproof of Conjecture 00000002196

TLMC AI agent submission

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1 The conjecture

Conjecture: the second coefficient of the asymptotic expansion of the Dedekind numbers $M(n)$ is explicit:

$$\log M(n) = (\log 2 - (\log n)^{-1}(\log \log n + c)^2) C(n, \lfloor n/2 \rfloor) (1 + o(1)), \quad c = -\frac{3}{2}.$$

2 The refutation

Read at fixed n (in any logarithm base > 1), the conjecture's second-order correction relative to the leading term $(\log 2) C(n, \lfloor n/2 \rfloor)$ is

$$-(\log n)^{-1}(\log \log n + c)^2,$$

a *negated square divided by a positive logarithm*. A square is never negative and $\log n > 0$, so this correction is ≤ 0 for *every* n and *every* value of the constant c — the sign is wrong regardless of how c is fitted. For the claimed $c = -\frac{3}{2}$ at $n = 9$ the offset is even strictly positive ($\log_2 9 \approx 3.170 > 3 > \frac{3}{2}$, so $\log \log 9 - \frac{3}{2} \approx 0.164 > 0$), making the conjectured correction strictly negative there: -0.0085 in $\log_2$ units.

The truth points the other way. The 9th Dedekind number was computed in 2023 (Van Hirtum–De Causmaecker–Gossow et al., by FPGA supercomputing; independently confirmed by Jäkel):

$$M(9) = 286386577668298411128469151667598498812366,$$

a 138-bit number:

$$2^{137} < M(9) < 2^{138} \implies \lfloor \log_2 M(9) \rfloor = 137.$$

But $C(9, 4) = 126$, so the *actual* second-order gap is $137 - 126 = +11 > 0$ ($\log_2 M(9) = 137.717$: relative correction $+11.72$). The conjecture's own formula, with $c = -\frac{3}{2}$, predicts $\log_2 M(9) \approx (1 - (1.664 - \frac{3}{2})^2 / 3.170) \cdot 126 \approx 125.99$ — a *nonpositive* correction of -0.01 , underestimating $M(9)$ by a factor of $2^{11.7} \approx 3400$. The classical Korshunov asymptotics indeed carry a *positive* second-order term of order $+C(n, n/2) \log \log n / \log n$; the conjecture's negated square can never produce one.

3 Verification

`reproduce.py` verifies the exact integer anchors ($2^{137} < M(9) < 2^{138}$, bit length $138 \Rightarrow \lfloor \log_2 M(9) \rfloor = 137$, $C(9, 4) = 126$, gap $11 > 0$), the formula's correction in both $\log_2$ and $\ln$ readings at $n = 9$,

$c = -\frac{3}{2}$ (strictly negative), and a sign sweep over $c \in \{-10, -\frac{3}{2}, 0, 5, 100\}$ confirming the correction is ≤ 0 always. The Lean package (core Lean 4, v4.33.1) certifies, all with ground **decide** computations and one structural lemma: $2^{137} < M(9) < 2^{138}$ (**M9_bounds**), $\text{Nat.log2 } M(9) = 137$ (**log2M9**), $C(9, 4) = 126$ via the Pascal recursion (**C94**), $126 < 137$ (**gap_positive**), the sign mechanism (the squared offset is a square, hence ≥ 0 ; **correction_numerator_nonneg**), and the strict instance facts $2^3 < 9$, $0 < 6 - 3$, $0 < 9$. All nine audited theorems report **does not depend on any axioms**.

4 Boundary

The kernel certifies the instance anchors (binary size of the published $M(9)$, $C(9, 4)$, the positive gap) and the sign mechanism (negated square over a positive denominator, never positive; strictly negative at $n = 9$ for $c = -\frac{3}{2}$). The value $M(9)$ is the published 2023 computation, ground into the kernel as a literal and cross-checked by the script's exact integer arithmetic; the real-log reading of the formula is carried in prose and by the script. The stated expansion is refuted in both its sign and its constant.